

VoiceRover

Tell a small robot car where to go, and it navigates there.

The Idea

The opening slides asked: “*What if you could drive a car with your phone?*” VoiceRover is a small robot car (Raspberry Pi + webcam, borrowed from MLH) that you control by speaking: “*Go to the red chair*”, “*Turn around and come back.*” Grok turns the request into driving steps, the webcam helps it find the target, and it talks back about what it’s doing. A tabletop prototype of the next era of vehicle interfaces.

Track Fit

Built with Cursor	Robot control code and web controller built in Cursor.
Grok Voice / Imagine	Voice commands and spoken status updates.
Space data	Not used. (Optional twist: frame it as a mini Mars rover.)
Navigation theme	Direct fit; matches a slide prompt.
Also eligible	Big Red, Hardware , Software.

Features

MVP

- Raspberry Pi drives two motors (forward, back, turn) from a web page on a phone.
- Voice command to a short list of actions to move the car.
- Live webcam view on the phone.
- Spoken confirmations: “*Moving forward one meter.*”

Stretch

- “Go to the red chair”: simple color or object detection to steer toward a target.
- Stop automatically for obstacles (ultrasonic distance sensor).
- Remember a route and replay it.

Tech Stack and Data

- Hardware: Raspberry Pi kit, Logitech webcam, motor driver, chassis, battery.
- Robot: Python (Flask/FastAPI) on the Pi; GPIO motor control.
- Vision: OpenCV color tracking (MVP), object detection (stretch).
- AI: Grok Voice to understand commands and reply.

Limitations and Risks

- **Parts availability.** MLH lists Raspberry Pis, Arduinos, webcams, and Grove parts, but *not* motors, a chassis, or batteries.
Mitigation: Check the MLH table first thing. Without wheels and motors, this idea isn't possible.
- **Hardware eats time.** Wiring, power problems, and flaky motors routinely cost hours.
Mitigation: Get the car driving from a keyboard by Saturday noon before adding voice.
- **Wi-Fi at the venue.** Phone-to-Pi control over crowded hackathon Wi-Fi can lag or drop.
Mitigation: Use the Pi as its own hotspot, or tether to a phone hotspot.
- **Voice to safe motion.** An AI misunderstanding could drive the car off a table.
Mitigation: Allow only a fixed set of safe actions with limits on distance and speed; add a stop button.
- **“Go to X” is hard.** Real object-seeking navigation is a research problem.
Mitigation: Use brightly colored targets for reliable detection.
- **Must return hardware.** Borrowed hardware must be returned after judging.
Mitigation: Plan for it; document the build with photos.

Scorecard

Demo-able	5/5	A moving robot at the table draws a crowd.
Buildable in 24h	3/5	Depends on parts and hardware luck.
Navigation theme	5/5	Driving with your phone and voice.
Wow factor	5/5	Physical and interactive.
Technical risk (5 = riskiest)	4/5	Hardware and parts.

Verdict

Best for a team with hardware experience. Can win both the Cursor and Hardware tracks, but confirm motors and a chassis exist before committing.